

Proposal Title:

Daily caffeine consumption and student self reported concentration during lecture

Research Questions:

- Is there a relationship between how much caffeine STA304H5 students consume each day and how well they can concentrate during lectures?
- Does this relationship still exist when we also consider students' sleep duration and stress levels?

Target Population:

Our target population is Fall 2026 STA304H5 students.

Sampling Method:

We will use stratified random sampling. STA304H5 students will be divided into two groups based on which lecture they are enrolled in (LEC0101 or LEC0102). We will then randomly select students from each lecture section so that both sections are represented in our sample.

Data Collection: After receiving feedback on our proposal and making any needed changes, we will share a Google Form questionnaire onto Piazza. The questions will ask students about the caffeine consumption, concentration, sleep duration, stress levels, and lecture section over the past seven days. We will aim for at least 60 usable responses before the October 9 deadline. To encourage participation, we will:

Offer to complete other groups' surveys.

Send one reminder through the same permitted channels.

There will be nine substantive questions and once course specific ID questions for participation credit and to prevent duplicate response. We will keep only the latest response for each ID. it will be a voluntary participation and no ID will be shown in the public results.

Simple Statistical Test:

We will use simple linear regression for research question 1 to examine the relationship between average daily caffeinated beverage consumption and self-reported concentration during lectures. Average daily consumption will be calculated by adding the number of servings

reported for coffee, tea, energy drinks, and other caffeinated beverages (Q3-Q6). The concentration score from Q7 will be the response variable.

Advanced Statistical Test:

Multiple linear regression with model selection. For research question 2, we will examine the relationship while also considering sleep duration and stress. We will compare models that include beverage consumption alone, with sleep, with stress, and with both. This will assist us in deciding which combination of variables to include. The results will show associations, but will not prove that caffeine causes changes in concentration.

Questionnaire Link:

https://docs.google.com/forms/d/e/1FAIpQLSdgsWaZUTG8jXkPriXbxz9i1uvidPZ_-F0VTWR9CfN1h9XTIq/viewform?usp=publish-editor